



Contents lists available at ScienceDirect

Science of the Total Environment

journal homepage: www.elsevier.com/locate/scitotenv

Paradigm shift in antibiotic-resistome of petroleum hydrocarbon contaminated soil

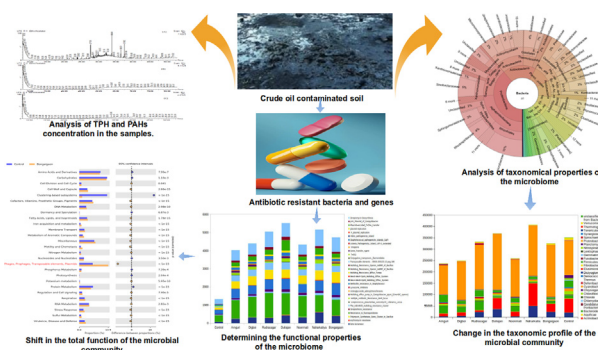
Nandita Das¹, Rhitu Kotoky¹, Anand Prakash Maurya, Bhriugu Bhuyan, Piyush Pandey^{*}

Department of Microbiology, Assam University, Silchar 788011, India

HIGHLIGHTS

- The emergence of antibiotic resistance from the environmental samples has been studied.
- The petroleum contaminated soil act as major reservoir of antibiotic resistance.
- The presence of PAHs and metals play significant role in development of resistome.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:

Received 21 August 2020

Received in revised form 29 October 2020

Accepted 29 October 2020

Available online xxxx

Editor: Ewa Korzeniewska

Keywords:

Metagenome
Resistome
Antibiotic
Environment
PAHs

ABSTRACT

The increasing prevalence of antibiotic-resistant microorganisms in both clinical and environmental samples is of great concern for public health. In the present study, environmental samples from seven different sites, heavily contaminated with petroleum hydrocarbons has been examined for the antimicrobial resistome through metagenomic approach. The soil samples were found to be contaminated with high concentration of total petroleum hydrocarbons (average 45 g/kg), polycyclic aromatic hydrocarbons (average $\sum 16\text{PAH} = 280 \text{ mg/kg}$), and heavy metals, which shapes the microbial community and their function. Proteobacteria was found to be predominant phylum in the contaminated habitat with the highest diversity (55.91%) followed by Actinobacteria (9.86%). Although the taxonomical abundance of the non-contaminated sample was not significantly different from contaminated samples, the functional abundance of genes related to antibiotic resistance was found to be higher up to 2 fold in contaminated samples. The comparative metagenomic analysis revealed a higher abundance of different antibiotic resistance genes, especially genes for fluoroquinolones was found to be higher up to 10 fold in contaminated samples. Moreover, the study has shown a significant difference in total functional diversity and abundance, mainly genes for aromatic compound metabolism and genes for phages, mobile genetic elements. These higher abundances of well recognized antibiotic resistance genes, multidrug efflux pumps, and integrons, suggest that the petroleum hydrocarbon contaminated sites can act as reservoirs for development of antibiotic resistance genes (ARGs). From this study, a significant link between the presence of petroleum hydrocarbon and the development of antibiotic resistance in the microbiome of contaminated habitat has been established.

© 2020 Elsevier B.V. All rights reserved.

1. Introduction

The continuous improvements of industrialization and urbanization level, the demand for petroleum oil, a nonrenewable energy source rises sharply worldwide. Thus the increase in usage have led to increase in contamination of soil with petroleum hydrocarbons, specially the oil drilling sites. Which have numerous effects on the soil biology and

^{*} Corresponding author.

E-mail address: piyushddn@gmail.com (P. Pandey).

¹ Contributed equally.

biochemistry, including development of environmental antibiotic resistance. The continuous development of antibiotic resistance is considered a major threat to mankind. Primarily it is believed that the antibiotic resistance is due to the large-scale use and misuse of antibiotics in humans and livestock which have led to the development of resistant to several, or even all known classes of clinically relevant antibiotics (Gorovtsov et al., 2018). However, the resistance is also readily detected in soil, aquatic, atmospheric, and other natural environments without exposure to clinical antibiotics that harbor microorganisms with multiple antibiotic resistant genes (ARGs) and capabilities for their mobilization (Surette and Wright, 2017). These resistant ARGs and their related genetic elements altogether present in a particular environment is referred to as “resistome” which is affected by many abiotic and biotic factors (Luzuriaga et al., 2012).

The soils are the largest reservoir of antibiotic-resistant bacteria (ARB) and ARGs due to the high diversity and density of microbial species present in there (Nesme and Simonet, 2015). Therefore, the resistome analysis of different soil samples provides new insight into environmental resistome, where different novel ARGs were frequently explored and discovered (Forsberg et al., 2014). The genetic and functional basis of the high diversity of environmental resistome may include different biotic and abiotic factors. The biotic factors include different secondary metabolites having antimicrobial activity produced by different microbes and plants and the abiotic factors include heavy metals (Baker-Austin et al., 2006), organic contaminants like polychlorinated biphenyls (Lo Giudice et al., 2013), pesticides (Anjum and Krakat, 2016), and disinfectants (Karatzas et al., 2007). These environmental factors participate widely in shaping bacterial communities and accordingly environmental resistome.

The resistome of an environment includes both intrinsic and acquired resistance mechanisms of microorganisms. The intrinsic resistance mechanisms include general cellular response to toxic molecules like broad-spectrum efflux pumps (Li et al., 2015a), chromosomally encoded resistant enzymes such as β -lactamases (Jacoby, 2009), and barriers to entry such as porins, and the outer membrane of gram-negative bacteria (Nikaido, 2003). Whereas the acquired resistance mechanism often obtained through horizontal gene transfer (HGT), including compound-specific efflux pumps, expression of non-sensitive targets, and enzymes that modify targets or the antibiotic molecules (Surette and Wright, 2017). All the mechanisms of a resistome may not be expressed and remain silent (Nazni et al., 2005), but the presence of resistance genes correlate with the potential antibiotic resistance driven by the genotype.

In this present study, we analyzed the metagenome profile of soil samples from seven different petroleum contaminated sites of Assam, India, and compared with a non-contaminated soil sample to study the shift in the resistome. The sample collection sites have been used as petroleum drilling sites for decades and so the microbial community of the sites was already adapted to the contamination of petroleum hydrocarbon and heavy metals and therefore supposed to become stable genetically.

2. Materials and methods

2.1. Collection of soil sample

The petroleum-contaminated soil samples were collected in November 2018 from heavily contaminated agricultural fields near seven different petroleum drilling sites of Assam, India. The sites were Amguri (26.8102° N, 94.5238° E), Bongaigaon (26.5030° N, 90.5536° E), Digboi (27.3951° N, 95.6300° E), Duliajan (27.3572° N, 95.3223° E), Naharkatiya (27.2870° N, 95.2476° E), Noonmati (26.1986° N, 91.8001° E), and Rudrasagar (26.9489° N, 94.5960°) (Supplementary file 1). The soil samples were collected aseptically and transported to the lab for sequencing at the low-temperature condition. The samples

were analyzed to determine the quantitative level of hydrocarbon and heavy metal contamination.

2.2. Quantitative estimation of TPH and PAHs in contaminated soil samples

The total petroleum hydrocarbons (TPH) and PAHs of the soil samples were quantified using Gas chromatography-Mass spectroscopy (GC-MS). TPH and PAHs were extracted using solvent extraction according to the US EPA 3545a procedure. Briefly, one gm of soil was collected and suspended in 25 ml of acetonitrile (ACN) and extracted for 15 min by ultrasonic stirring and then centrifuged at 10000 rpm for 12 mins. Aliquots of 1 ml of each sample were diluted in 10 ml ACN and analyzed using GC-MS through methods as described by Kotoky and Pandey (2020a).

2.3. DNA extraction and metagenome analysis

Metagenomic DNA from crude oil contaminated sites was independently extracted using the Nucleospin Soil DNA isolation Kit. Paired-end library preparation of QC passed DNA samples were done using TruSeq Nano DNA Library Prep Kit. After obtaining desired size paired-end libraries, loaded onto a NextSeq 500 Illumina platform for cluster generation and sequencing. After sequencing, raw data were processed using Trimmomatic v 0.35 to obtain high-quality clean reads, where adaptor sequences, chimeras, ambiguous reads (reads with an unknown nucleotide), and low-quality reads (reads with more than 10% quality threshold (QC) < 20 Phred score) were removed. Moreover, the meta-genome sequencing data set of non-contaminated agriculture soil samples (MGM4821297.3), from geographic location (latitude 25.4255° N, longitude 84.1627° E), which has not been exposed to any crude oil contamination, was retrieved from the MG-RAST database, served as control for the comparison of the resistome.

The selection of control site had been crucial for this study as it may be argued that there may be natural variation in the bacterial diversity of the control and contaminated soils, and also the diversity due to the geographical variations. In fact, different studies have investigated the spatial patterns in microbial biodiversity at regional, continental (Lauber et al., 2009), and global scales (Tedersoo et al., 2012). However, the difference in biogeographic pattern depends on several underlined mechanisms with limited understanding (Nemergut et al., 2013; Wang et al., 2013), which also includes the environmental factors and the contaminants. According to Vellend's (2010) conceptual synthesis theory, selection is one of the most effective methods that contribute to the composition and diversity of species in a particular niche. Selection is the major factor that results from the environmental pressures causing variation in survival and pattern of population diversity. As the study area is under heavy hydrocarbon related activities, it was imperative to have control from a site, having no history of hydrocarbon contamination and have similar climatic and edaphic conditions. In fact, the control site is located at similar latitude (24–26 N) and height from sea level (ICAR, 2019), and also, the sites for soil sampling, including control, comes under similar soil topography, (alluvial soil), as classified for India (Bhattacharyya et al., 2013). Therefore, in case of present study it may be considered that the difference in microbial community between the samples is mainly due to the selection pressure due to petroleum hydrocarbons contamination.

2.4. Bioinformatics analysis

Taxonomic analysis of the predicted genes of samples was first carried out using the Kaiju database. Also, the raw data obtained from the Illumina sequencing platform were uploaded in Metagenome Rapid Annotation using Subsystem Technology (MG-RAST) to remove uninformative and duplicative reads for further analysis. Additionally, the affiliation of functional genes to the taxonomic community was analyzed through KO (KEGG orthology), NOG (non-supervised orthologous

groups), COG (cluster of orthologous groups) and Subsystem sources using the best hit classification system. The abundance (number of sequences) of the ARG were analyzed and compared subsequently. The comparative analysis of the functional abundance between the samples was done and visualized through Statistical Analysis of Metagenomic profile (STAMP).

The abundance of ARGs along with taxonomic abundance was analyzed through BusyBee where the input sequences are taxonomically and functionally annotated and represented as individual points in the 2D scatter plot. Convex hull delineates the predicted cluster whereas the antibiotic resistance genes are represented as points of larger size and dark color.

3. Results & discussion

3.1. Concentration of PAHs in the samples

The concentration of total petroleum hydrocarbons (TPH) and PAHs in the samples were measured with GC-MS analysis. The average concentrations of $\Sigma 16$ PAH in the samples were ranged from 0.780 mg/kg to 560.05 mg/kg which includes different high molecular weight PAHs like pyrene, anthracene, benzo(a)pyrene. The concentration of total petroleum hydrocarbons (TPH) was also found to be significantly high. The highest concentration of TPH was found to be 75 g/kg of surface soil in the contaminated area of Duliajan whereas the lowest concentration of TPH was 1109 mg/kg of soil in Bongaigaon samples, which was significantly higher than normal level of contamination reported.

The petroleum hydrocarbons are hydrophobic in the environment and making strong links with non-polar hydrophobic moieties in contaminated soils (Bengtsson-Palme et al., 2018; Chen et al., 2019; Rodgers et al., 2019), therefore, it became challenging to measure the definite level of hydrocarbons in the contaminated areas. However, in the present study, the determined concentration of PAHs and TPHs was significantly high in the contaminated samples. Soil is the primary steady reservoir for PAHs as they are readily absorbed by organic matter. Which leads to the accumulation of PAHs in food chains, which could cause a potential risk to human health (Yang et al., 2014).

3.2. Taxonomic profiling of the antibiotic resistance bacteria (ARB) of the contaminated sites

The overall taxonomic profiling of the samples was done along with the bacterial groups with antibiotic resistance. The bacterial group with resistance genes were considered ARB although have not been confirmed phenotypically. At phylum level of overall taxonomic profiling, Proteobacteria was found to be predominant with highest diversity (53–58%) followed by Actinobacteria (8–10%), Bacteroides (5–7%), Acidobacteria (4–6%), Firmicutes (3–5%), Chloroflexi (2–4%), Cyanobacteria (2–4%), and Euryarchaeota (1.5–4%) (Fig. 1A). The analysis also revealed several groups of bacterial species in the contaminated sites related to a different group of antibiotic resistance (Fig. 1B). The most common bacterial species group present in the contaminated sites and their related resistance property are given below (Fig. 2). The comparative analysis showed that the contamination of petroleum oil has no significant impact on the abundance of bacterial groups but have a huge impact on the bacterial groups related to antibiotic resistance. It has demonstrated that the contamination has led to the replacement of many bacterial groups with the antibiotic resistance groups or have developed the resistance within. It was noted that the level of hydrocarbon contamination was not significantly co-related with the bacterial population quantitatively, however the functional properties of the bacterial population strongly correlated with the level of contamination. The bacterial functions related to antibiotic resistance was found to be increasing with the contamination of hydrocarbons. From the analysis it can be assumed that the crude oil

contamination is one of the major factors that determines the microbiome of soil, specially the functional attributes (the aspect of the present study).

In the petroleum-contaminated samples, the presence of several groups of bacteria resistance to beta-lactam, fluoroquinolones, erythromycin, vancomycin, fosfomycin has indicated that crude oil-contaminated areas are acting as a reservoir for antibiotic resistance and transfer. Ben Said et al. (2008) also reported several bacterial species that were involved in the degradation of PAHs compounds and found to be resistant to a wide range of antibiotics. Similarly, different PAH degrading bacterial isolates *P. fluorescens* BBN1, *P. mandelii* CSGN1, *P. corrugata* BBB2, and *P. syringae* CSGN4 were found to be resistant to amoxicillin, piperacillin, piperacillin-tazobactam, expanded spectrum cephalosporins, moxalactam, novobiocin, and tigecycline (Mathe et al., 2012). These results suggest that the presence of PAHs may influence the development of antibiotic resistance in the microbiome.

3.3. Functional profiling of the antibiotic resistome of the contaminated samples

The functional profiling of the resistome of the samples was analyzed that revealed occurrence of different intrinsic and extrinsic resistance mechanisms. The analysis showed the presence of ARGs of multiple types, efflux protein systems, and different integrons and mobile genetic elements. The heat map of all the resistant mechanisms and their genetic basis in the contaminated sites showed that the mechanisms including acroflavin resistant proteins, fluoroquinolones resistance through DNA gyrase subunits, Beta-lactamase were found to be most abundant and diverse (Fig. 3).

3.3.1. The occurrence of antibiotic resistance genes in the contaminated samples

The metagenomic analysis revealed occurrences of different antibiotic resistance genes of different types of conferring resistance to multiple groups of antibiotics from seven different contaminated sites. Among these, twenty-four different types of ARGs including beta-lactamase genes of Class A (penicillinases), B (metallo-hydrolases/metallo-beta-lactamases), C (AmpC) and D (oxacillinases), methicillin resistance staphylococci, erythromycin, vancomycin, and fosfomycin were found to be present in the samples (Fig. 4). For the analysis the abundance (number of gene sequences) were determined from the functional abundance profile and compared. Among all the seven sites of crude oil contamination, beta-lactamase resistance genes were most predominant (41.0%), followed by fluoroquinolones (26.07%) and erythromycin resistance genes (25.0%) and have a significant difference in comparison to the control. The abundance of genes for fluoroquinolones was found higher up to 10 folds in the contaminated samples than in non-contaminated control whereas the genes for methicillin resistance were 5 folds higher.

Several earlier studies also reported antibiotic resistance related to environmental contamination analyzed through MG-RAST tool (Yadav and Kapley, 2019; Imchen et al., 2018). The MG-RAST analysis uses an automated analysis pipeline through the M5nr (MD5- based non-redundant protein database) for annotation (Wilke et al., 2012). The M5nr is an integration of many sequence databases into one single database comprising several databases like European Bioinformatics Institute (EBI), Gene Ontology (GO), Joint Genome Institute (JGI), Kyoto Encyclopedia of Genes and Genomes (KEGG), Genbank of National Center for Biotechnology Information (NCBI), Phage Annotation Tools and Methods (Phantome), SEED subsystem, UniProt, Virginia Bioinformatics Institute (VBI), evolutionary genealogy of genes (eggNOG), that compute the annotations of the metagenome sequences into functions including the antibiotic resistance genes (Keegan et al., 2016). Moreover, the groups of bacteria related to particular resistance has been recognized through the analysis.

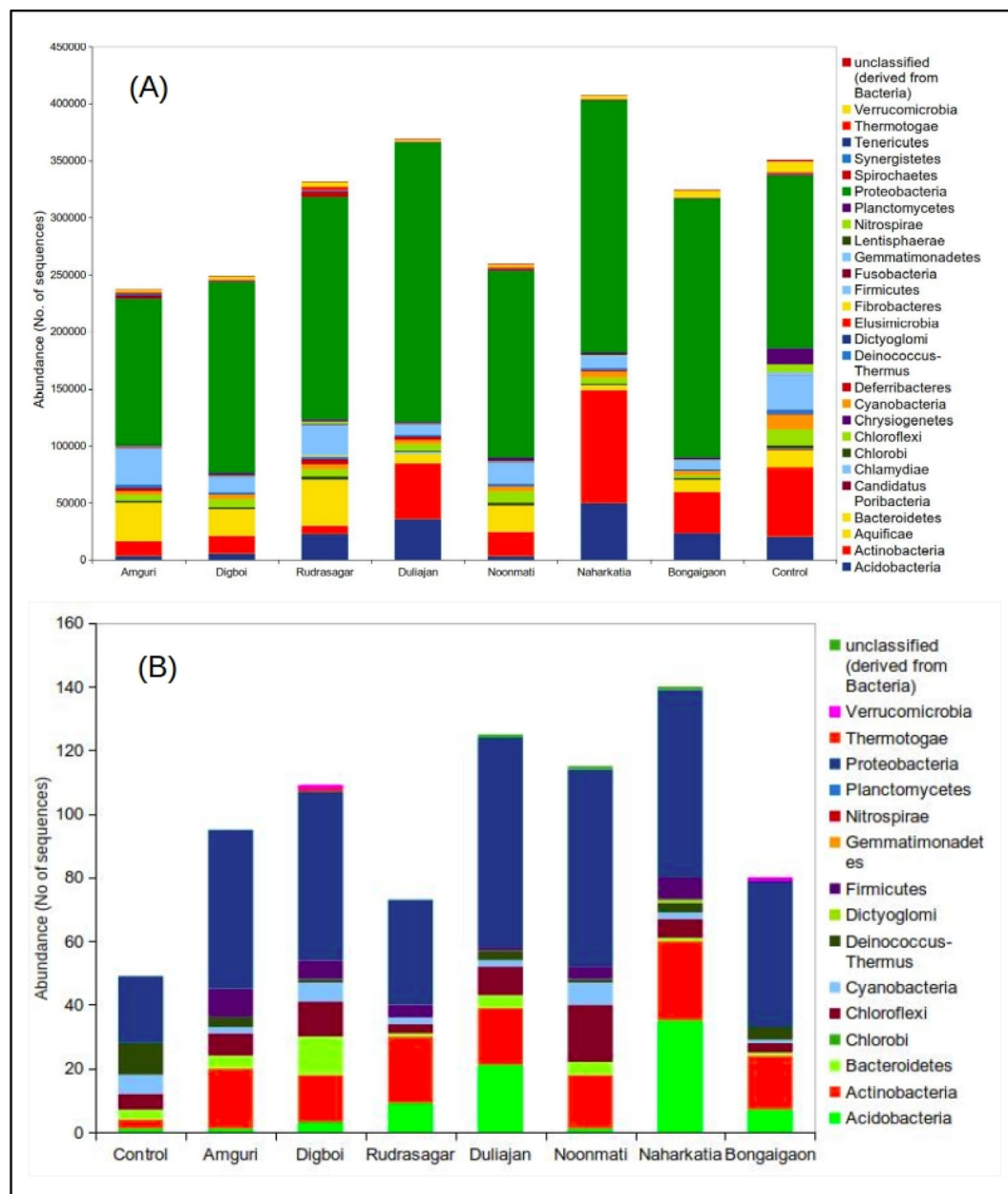


Fig. 1. Abundance of total bacterial phylum in the seven crude oil contaminated areas compared with control (A) and bacterial groups related to antibiotic resistance (B).

In an interesting study, Li et al. (2015b), demonstrated co-occurrence of environmental antibiotic genes in different environmental samples, analyzed through metagenomic approach. In that study it was observed that the abundance of total ARG in environments were strongly co-related with the levels of anthropogenic impacts, where in the high impacted environments the total ARG abundances increased up to three orders of magnitude than the less impacted environments. Similarly, in the present study also the abundance of ARGs were found to be much higher in the crude oil contaminated soil than non-contaminated soil sample.

Earlier studies also suggested that the contaminated areas like activated sludge, metal-contaminated habitats which cause abiotic stress on the microbiome may serve as hotspot exchange of resistant genes due to different selective pressure (Li et al., 2015a; Luo et al., 2016).

Hemala et al. (2014) also demonstrated an association between the antimicrobial resistance and crude oils contamination, where the hydrocarbon-degrading Gammaproteobacteria and Actinobacteria were found to harbor multiple antibiotics resistance genes. Similarly, Chen et al. (2016) reported the presence of penicillin, cefoxitin, cefazolin, ceftriaxone, chloramphenicol, aminoglycosides, macrolides, fluoroquinolones, tetracyclines, polypeptides resistance genes in a PAHs contaminated habitat. Apart from the antimicrobial resistance in the whole microbiome, different other studies also demonstrated the antimicrobial resistance in culturable bacteria with hydrocarbon-degrading capabilities, isolated from petroleum contaminated sites. Therefore, it may be considered that crude oil contamination is involved in elevating ARBs and ARGs by dissemination through horizontal or vertical transmission mechanisms.

Fig. 2. A bubble diagram showing bacterial sp. involved in different antimicrobial activities in different samples. In this diagram colour variation indicates different sites and size of the bubble represents the abundance of respective bacteria.



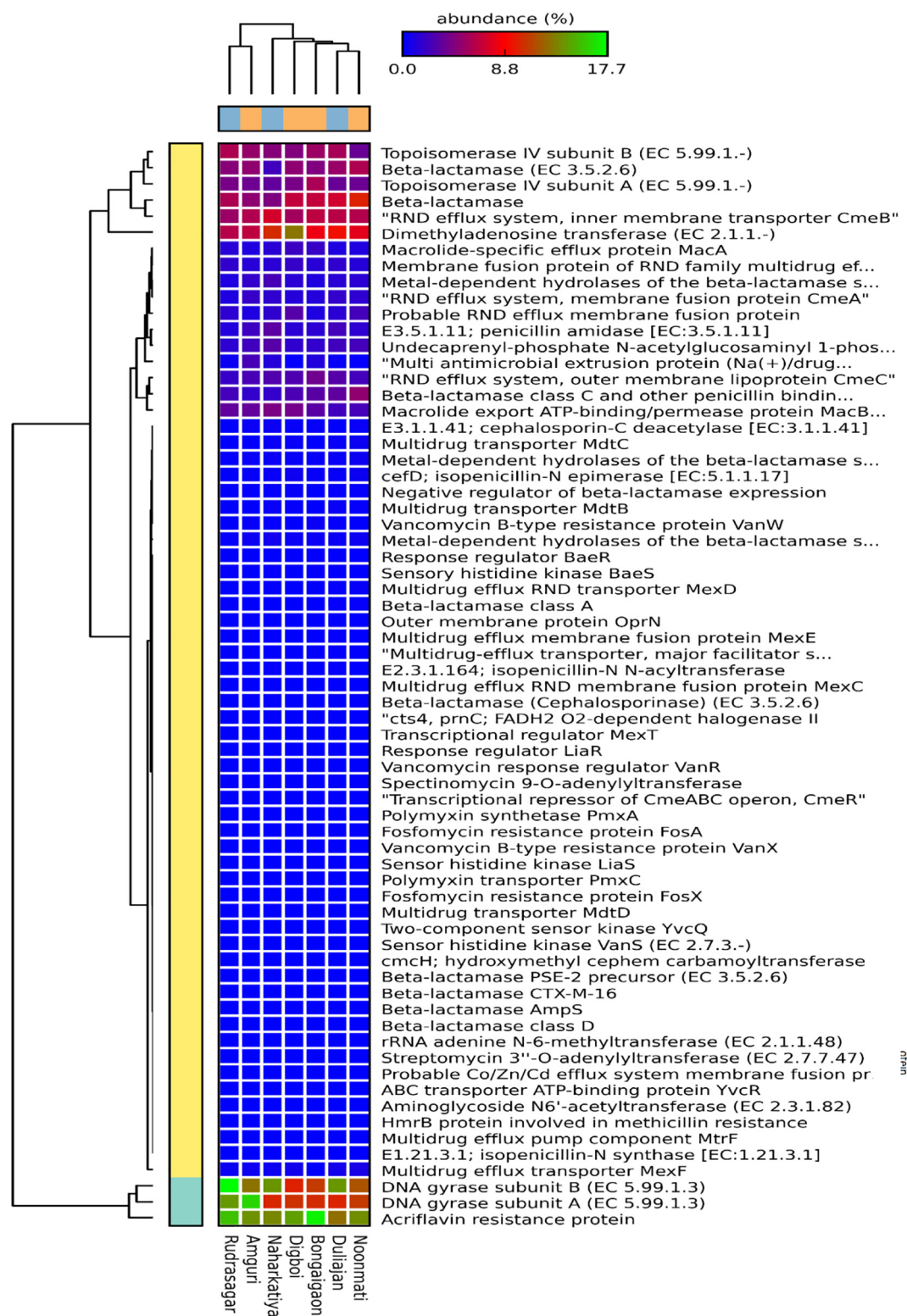


Fig. 3. Heat map of all the antibiotic resistant mechanisms and their genetic basis in the contaminated sites.

3.3.2. Occurrence of antibiotic efflux pumps in the contaminated samples

The analysis demonstrated the presence of four different groups of efflux pumps including two different multi-drug resistant (MDR) efflux pump, MDR efflux of *Campylobacter jejuni* (CmeABC operon) and MexE-MexF-OprN MDR efflux systems in the petroleum-contaminated samples. Among these four groups, 33 different types of specific efflux

pumps were found comprising the membrane fusion protein of RND, MexC and MexE efflux, multidrug transporter protein MdtB, MdtC, mdtD, MexD, MexE, MexF and MFS, macrolide resistance specific efflux, efflux protein MacA, MacB, acriflavin resistance protein, response regulator BaeR, sensory histidine kinase BaeS, membrane transporter CmeA, CmeB, CmeC, CmeR, macrolide-specific efflux protein MacA,

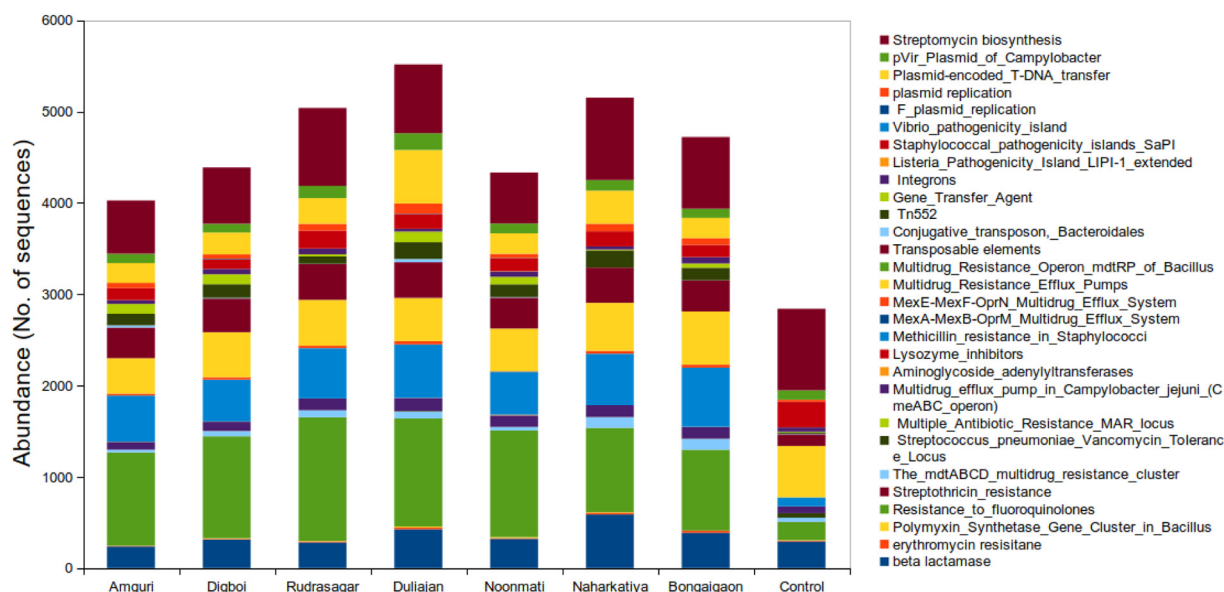


Fig. 4. Abundance of ARGs in contaminated soils.

transcriptional regulator *MexT*, and outer membrane protein *OprN* were present. In our study, MDR efflux pump was the most predominant (69–72%) followed by MDR efflux of *Campylobacter jejuni* (*CmeABC*

operon) (26–30%) among all the groups of efflux pump found in the samples (Supplementary file 2).

The presence of genes for efflux pumps like *RND*, *MexC*, *MexE*, *MdtB*, *MdtC*, *mdtD*, *MexD*, *MexE*, *MexF*, *MFS*, *MacB*, *BaeR*, *BaeS*, *CmeA*, *CmeB*, *CmeC*, *CmeR*, *MacA*, *MexT*, and *OprN* etc. has been known to involve in enrichment of ARGs in a microbiome. The *RND* efflux pumps are presumed to be responsible for the expression of multidrug resistance, and environmental adaptability (Martinez et al., 2009), as well as degradation of different PAHs (Hearn et al., 2003; Chain et al., 2006). Efflux protein *EmhABC* of *P. fluorescens* LP6a was reported to be involved in the discharging of antibiotics and PAHs (Hearn et al., 2003; Adebusi and Foght, 2011). The MDR efflux systems *MexE-MexF-OprN*, are tripartite efflux pumps which encoded in operons of *MexEF-OprN* are reported to be responsible in quorum sensing (QS) and also release of toxic metabolites as documented in *P. aeruginosa* (Strateva and Yordanov, 2009). The two-component system *czcRS* that are known to regulate *czc* efflux pump (for heavy metal resistance) also reported regulating porins protein *OprD* responsible for resistance to imipenem in *P. aeruginosa* (Liu et al., 2015). The high abundance of class 1 integron (97%) in the contaminated habitat suggest a high rate of transfer of ARGs in the environment. It has been found that genes encoding the MDR efflux pump usually located in class 1 integrons that have a connecting link between PAH contamination and enrichment of ARGs. The class 1 integron borne gene transfer was primarily responsible for a greater abundance of ARGs in soil microbiome (Hutchinson et al., 2004). The presence of conjugative transposon, *Tn552*, *ISPsy4*, putative transposases, *F* plasmids, and plasmid partition proteins supports the HGT that helps bacteria to adapt and evolve in the different environments by acquiring the resistance genes.

3.3.3. Role of integrons and other mobile genetic elements (MGEs)

Integrons are the mobile elements which are present in the environment and helps bacteria to exchange their genes. The analysis of the samples showed a high abundance of class 1 integron (97%) in all the crude oil-contaminated areas while class 2 was 1.22% and class 4 integrons were 1.84%. Besides the presence of integrons, there were transposable elements *Tn552*, *ISPsy4*, putative transposases: *TniA* *TniB*, *TniQ*, Transposase *OrfAB*, Conjugative transposon proteins: *TraA*, *TraBm* *TraE*, *TraF*, *TraG*, *TraJ*, *TraK*, *TraM*, *TraN*, *TraO*, *TraQ*, plasmid partition proteins: *ParA*, *ParB*, and *ParB-2* were also present in the samples (Supplementary file 3). These transposable elements involved in exchanging

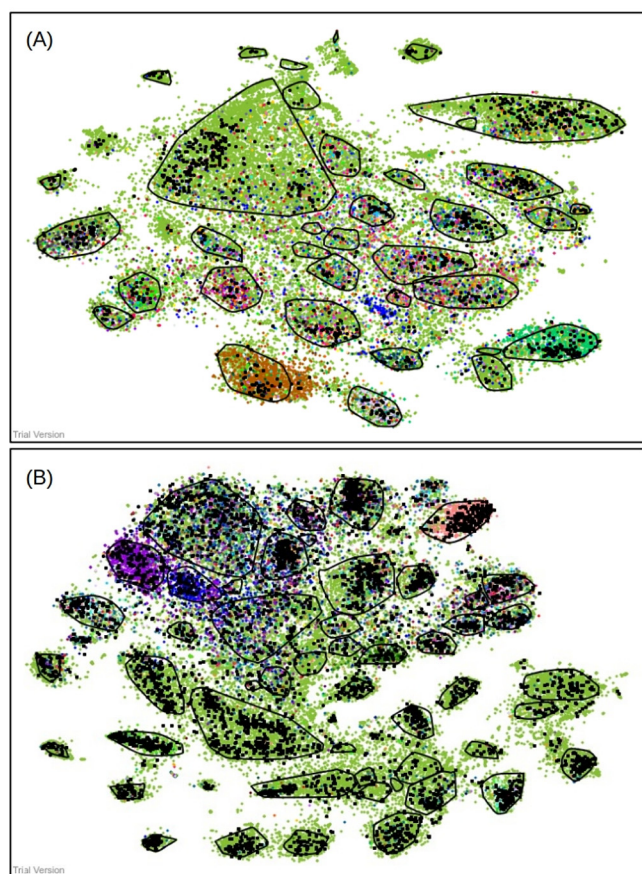


Fig. 5. Taxonomic (upto genus level) and functional diversity (ARGs) present in the control sample (A) and petroleum contaminated sample (B). Colored dots indicate abundance of microbes and the black dots the presence of ARGs. Convex hulls (black polygons) delineate the predicted clusters that occurs closely together.

the resistance genes and integron gene cassettes to other bacteria. Conjugative transposon and F plasmid support to ARBs in the conjugative transfer/horizontal genes transmission (HGT) of ARGs to bacteria in multiple stress conditions of PAHs in to the crude oil-contaminated areas.

3.4. Analysis through BusyBee

The comparative analysis of the abundance of ARGs in control and petroleum-contaminated soil samples has also been done through BusyBee and plotted as a scatter plot (Fig. 5a and b). The metagenomic analysis of the samples using BusyBee provided a wider insight into the resistome analysis as it analyze coding sequences against ResFams (collection of antibiotic resistance genes) and thereby can detect a wide range of ARGs. The taxonomical and functional annotations of the two samples: one uncontaminated control sample and one contaminated sample (Amguri) has been presented in Fig. 5. Where each color dot represents an individual sequence colored taxonomically at the genus level and the bold black dots represent the ARGs genes. It has been observed that the microbial taxonomic abundance of the samples did not differ much but the abundance of ARGs was significant.

3.5. Difference in total functional abundance

The comparative analysis of functional abundance was analyzed where the abundance of genes categorized into different SEED subsystem was compared between samples of control and contaminated samples and visualized through STAMP. The analysis showed a higher abundance of genes related to phages, prophages, transposable elements, and plasmids in the contaminated samples than in non-contaminated samples (Fig. 6). Moreover, higher abundance was also observed for the genes related to aromatic compound degradation, stress response, virulence, motility, and chemotaxis, etc. at the contaminated soil samples. However, the genes related to protein metabolism, and carbohydrate metabolism were found less in contaminated samples.

These analyses and the results demonstrate a clear link between contamination of PAHs and the development of resistome and antibiotic resistance of microbiome not only related to clinical exposure of antibiotics. Moreover, the contamination led to a change in the functional diversity and abundance of the microbiome of the contaminated soils as we reported previously (Kotoky and Pandey, 2020b). Several other earlier metagenomic analyses have also reported that the antibiotic resistance genes are ubiquitous in a variety of ecosystem including pristine

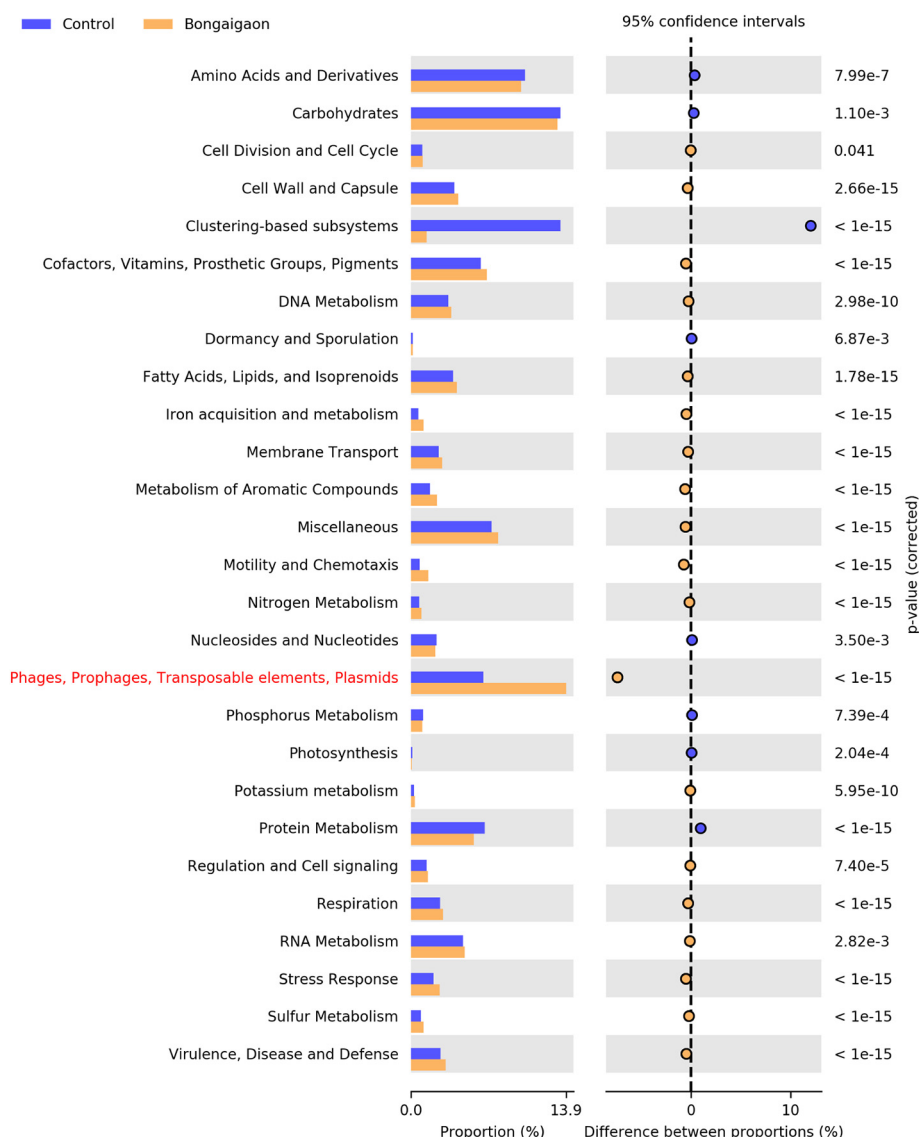


Fig. 6. Difference of functional genes categorized into different subsystem between contaminated and non-contaminated (control) samples.

environments (Imchen et al., 2018; Zeng et al., 2019). Recently, Cunningham et al. (2020) reviewed the potential risk of antibiotic resistant bacteria and genes in petroleum contaminated sites and in the process of bioremediation and demonstrated how the antibiotic resistant bacteria may become more abundant in contaminated soils due to co-selection pressures from pollutants such as metals and hydrocarbons. Yadav and Kapley (2019) used a similar approach to study the resistome of activated sludge. However, the activated sludge process is commonly used during the treatment of industrial and domestic effluents that contain toxic pollutants including disposed of antibiotics. Therefore, the presence of resistome as found in the study could be anticipated.

4. Conclusion

In the present study, the heterogeneous hydrocarbon contamination has been studied in a natural environment for antibiotic resistome, including functional genes. Further, the relationship of taxonomical abundance with the resistome of a given sample is still unclear, as it is considered that the resistome is mostly factor-driven rather than driven by a particular group of microorganisms. Therefore, the data provided in the present work provides strong evidence that the petroleum hydrocarbon contaminated sites could act as reservoirs for ARGs with indigenous molecular mechanisms available for their development and genetic transfers.

Data availability

The data of metagenomic analysis of the crude oil-contaminated field soil was registered in NCBI under ID SUB4517618 and SUB4522993, BioSample accession of the samples SAMN10068252, SAMN10068251, SAMN10068250, SAMN10068248, SAMN10068247, SAMN10068246, and BioProject ID PRJNA491236. Metagenomic sequences deposited in the MG-RAST system having deposition ID 4832982.3, 4832983.3, 4832984.3, 4832985.3, 4832986.3, 4832987.3 and 4880959.3 respectively.

CRedit authorship contribution statement

ND has done the sample collection and formal analysis. RK has done the acquisition, analysis, and interpretation of the data and written the draft manuscript. APM has done the formal analysis, acquisition of the data. BB has done the GCMS analysis of TPH concentration. PP has conceptualized and designed the work and also prepared, reviewed, and finalized the manuscript. ND and RK have equal contributions in the manuscript.

Declaration of competing interest

The authors declare no competing interest.

Acknowledgment

ND, APM, BB, and PP acknowledges the Department of Biotechnology (DBT), Govt. of India for financial support, RK acknowledges the Department of Science and Technology (DST), Govt. of India for DST-INSPIRE fellowship.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jscitotenv.2020.143777>.

References

Adebusuyi, A.A., Foght, J.M., 2011. An alternative physiological role for the EmhABC efflux pump in *Pseudomonas fluorescens* cLP6a. *BMC Microbiol.* 11, 252. <https://doi.org/10.1186/1471-2180-11-252>.

- Anjum, R., Krakat, N., 2016. Detection of multiple resistances, biofilm formation and conjugative transfer of *Bacillus cereus* from contaminated soils. *Curr. Microbiol.* 72, 321–328.
- Baker-Austin, C., Wright, M.S., Stepanauskas, R., McArthur, J.V., 2006. Co-selection of antibiotic and metal resistance. *Trends Microbiol.* 14 (4), 176–182.
- Ben Said, O., Goñi-Urriza, M.S., El Bour, M., et al., 2008. Characterization of aerobic polycyclic aromatic hydrocarbon-degrading bacteria from Bizerte lagoon sediments, Tunisia. *J. Appl. Microbiol.* 104, 987–997. <https://doi.org/10.1111/j.1365-2672.2007.03621.x>.
- Bengtsson-Palme, J., Kristiansson, E., Larsson, D.G.J., 2018. Environmental factors influencing the development and spread of antibiotic resistance. *FEMS Microbiol. Rev.* 53, 68–80. <https://doi.org/10.1093/femsre/fux053>.
- Bhattacharyya, T., Pal, D., Mandal, C., Chandran, P., Ray, S., Sarkar, D., Velmourougane, K., Srivastava, A., Sidhu, G.S., Singh, S., Sahoo, A., Dutta, D., Nair, K.M., Srivastava, R., Tiwary, P., Nagar, A.P., Nimkhedkar, S.S., 2013. Soils of India: historical perspective, classification and recent advances. *Curr. Sci.* 104, 1308–1323.
- Chain, P.S.G., Deneff, V.J.K., Konstantinidis, T., et al., 2006. *Burkholderia xenovorans* LB400 harbors a multi-replicon, 9.73-Mbp genome shaped for versatility. *Proc. Natl. Acad. Sci.* 103, 15280–15287. <https://doi.org/10.1073/pnas.0606924103>.
- Chen, B., He, R., Yuan, K., et al., 2016. Polycyclic aromatic hydrocarbons (PAHs) enriching antibiotic resistance genes (ARGs) in the soils. *Environ. Pollut.* 220, 1005–1013. <https://doi.org/10.1016/j.envpol.2016.11.047>.
- Chen, Y., Su, J.Q., Zhang, J., et al., 2019. High-throughput profiling of antibiotic resistance gene dynamic in a drinking water river-reservoir system. *Water Res.* 149, 179–189. <https://doi.org/10.1016/j.watres.2018.11.007>.
- Cunningham, C.J., Kuyukina, M.S., Ivshina, I.B., Konev, A.I., Peshkur, T.A., Knapp, C.W., 2020. Potential risks of antibiotic resistant bacteria and genes in bioremediation of petroleum hydrocarbon contaminated soils. *Environ Sci Process Impacts* 22 (5), 1110–1124.
- Forsberg, K.J., Patel, S., Gibson, M.K., Lauber, C.L., Knight, R., Fierer, N., Dantas, G., 2014. Bacterial phylogeny structures soil resistomes across habitats. *Nature* 509, 612–616.
- Gorovtsov, A.V., Sazykin, I.S., Sazykina, M.A., 2018. The influence of heavy metals, polyaromatic hydrocarbons, and polychlorinated biphenyls pollution on the development of antibiotic resistance in soils. *Environ. Sci. Pollut. Res.* 25, 9283–9292. <https://doi.org/10.1007/s11356-018-1465-9>.
- Hearn, E.M., Dennis, J.J., Gray, M.R., Foght, J.M., 2003. Identification and characterization of the emhABC efflux system for polycyclic aromatic hydrocarbons in *Pseudomonas fluorescens* cLP6a. *J. Bacteriol.* 185, 6233–6240. <https://doi.org/10.1128/JB.185.21.6233-6240.2003>.
- Hemala, L., Zhang, D., Margesin, R., 2014. Cold-active antibacterial and antifungal activities and antibiotic resistance of bacteria isolated from an alpine hydrocarbon-contaminated industrial site. *Res. Microbiol.* 165, 447–456. <https://doi.org/10.1016/j.resmic.2014.05.035>.
- Hutchinson, M.L., Walters, L.D., Moore, A., et al., 2004. Effect of length of time before incorporation on survival of pathogenic bacteria present in livestock wastes applied to agricultural soil. *Appl. Environ. Microbiol.* 70, 5111–5118. <https://doi.org/10.1128/AEM.70.9.5111-5118.2004>.
- ICAR, 2019. *Annual Report, 2019–2020*.
- Imchen, M., Kumavath, R., Barh, D., Vaz, A., Góes-Neto, A., Tiwari, S., Ghosh, P., Wattam, A.R., Azevedo, V., 2018. Comparative mangrove metagenome reveals global prevalence of heavy metals and antibiotic resistome across different ecosystems. *Sci. Rep.* 8 (1), 11187. <https://doi.org/10.1038/s41598-018-29521-4>.
- Jacoby, G.A., 2009. AmpC β -lactamases. *Clin. Microbiol. Rev.* 22 (1), 161–182.
- Karatzas, K.A., Webber, M.A., Jorgensen, F., Woodward, M.J., Piddock, L.J., Humphrey, T.J., 2007. Prolonged treatment of *Salmonella enterica* serovar Typhimurium with commercial disinfectants selects for multiple antibiotic resistance, increased efflux and reduced invasiveness. *J. Antimicrob. Chemother.* 60, 947–955.
- Keegan, K.P., Glass, E.M., Meyer, F., 2016. MGRAST, a metagenomics service for analysis of microbial community structure and function. In: Martin, F., Uroz, S. (Eds.), *Microbial Environmental Genomics (MEG). Methods in Molecular Biology*. Humana Press, New York, NY, p. 1399.
- Kotoky, R., Pandey, P., 2020a. Rhizosphere assisted biodegradation of benzo(a)pyrene by cadmium resistant plant-probiotic *Serratia marcescens* S217, and its genomic traits. *Sci. Rep.* 10, 5279. <https://doi.org/10.1038/s41598-020-62285-4>.
- Kotoky, R., Pandey, P., 2020b. Difference in the rhizosphere microbiome of *Melia azedarach* during removal of benzo(a)pyrene from cadmium co-contaminated soil. *Chemosphere* 258, 127175. <https://doi.org/10.1016/j.chemosphere.2020.127175> epub ahead of print, 2020 May 28.
- Lauber, C.L., Hamady, M., Knight, R., Fierer, N., 2009. Pyrosequencing-based assessment of soil pH as a predictor of soil bacterial community structure at the continental scale. *Appl. Environ. Microbiol.* 75, 5111–5120.
- Li, X.Z., Plésiat, P., Nikaido, H., 2015a. The challenge of efflux-mediated antibiotic resistance in gram-negative bacteria. *Clin. Microbiol. Rev.* 28 (2), 337–418.
- Li, B., Yang, Y., Ma, L., Ju, F., Guo, F., Tiedje, J.M., Zhang, T., 2015b. Metagenomic and network analysis reveal wide distribution and co-occurrence of environmental antibiotic resistance genes. *The ISME journal* 9 (11), 2490–2502.
- Liu, P., Chen, X., Huang, Q., Chen, W., 2015. The role of CzcRS two-component systems in the heavy metal resistance of *Pseudomonas putida* X4. *Int. J. Mol. Sci.* 16 (8), 17005–17017. <https://doi.org/10.3390/ijms160817005>.
- Lo Giudice, A., Casella, P., Bruni, V., Michaud, L., 2013. Response of bacterial isolates from Antarctic shallow sediments towards heavy metals, antibiotics and polychlorinated biphenyls. *Ecotoxicology* 22, 240–250.
- Luo, W., Xu, Z., Riber, L., Hansen, L.H., 2016. Diverse gene functions in a soil mobilome. *Soil Biol. Biochem.* 101, 175–183.
- Luzuriaga, A.L., Sánchez, A.M., Maestre, F.T., Escudero, A., 2012. Assemblage of a semi-arid annual plant community: abiotic and biotic filters act hierarchically. *PLoS One* 7, 1–9.

- Martínez, J.L., Sánchez, M.B., Martínez-Solano, L., et al., 2009. Functional role of bacterial multidrug efflux pumps in microbial natural ecosystems. *FEMS Microbiol. Rev.* 33, 430–449.
- Mathe, I., Benedek, T., Tancsics, A., Palatinszky, M., Lanyi, S., Marialigeti, K., 2012. Diversity, activity, antibiotic and heavy metal resistance of bacteria from petroleum hydrocarbon contaminated soils located in Harghita County (Romania). *Int. Biodeterior. Biodegr.* 73, 41–49.
- Nazni, W.A., Luke, H., Wan Rozita, W.M., Abdullah, A.G., Sadiyah, I., et al., 2005. Determination of the flight range and dispersal of the house fly, *Musca domestica* (L.) using mark release recapture technique. *Trop. Biomed.* 22 (1), 53–61.
- Nemergut, D.R., Schmidt, S.K., Fukami, T., O'Neill, S.P., Bilinski, T.M., Stanish, L.F., et al., 2013. Patterns and processes of microbial community assembly. *Microbiol. Mol. Biol. Rev.* 77, 342–356.
- Nesme, J., Simonet, P., 2015. The soil resistome: a critical review on antibiotic resistance origins, ecology, and dissemination potential in telluric bacteria. *Environ. Microbiol.* 17, 913–930.
- Nikaido, H., 2003. Molecular basis of bacterial outer membrane permeability revisited. *Microbiol. Mol. Biol. Rev.* 67 (4), 593–656.
- Rodgers, K., McLellan, I., Peshkur, T., et al., 2019. Can the legacy of industrial pollution influence antimicrobial resistance in estuarine sediments? *Environ. Chem. Lett.* 17, 595–607.
- Strateva, T., Yordanov, D., 2009. *Pseudomonas aeruginosa* - a phenomenon of bacterial resistance. *J. Med. Microbiol.* 58, 1133–1148.
- Surette, M., Wright, G.D., 2017. Lessons from environmental antibiotic resistome. *Ann Rev Microbiol* 71 (1), 309–329. <https://doi.org/10.1146/annurev-micro-090816-093420>.
- Tedersoo, L., Bahram, M., Toots, M., Diedhiou, A.G., Henkel, T.W., Kjoller, R., et al., 2012. Towards global patterns in the diversity and community structure of ectomycorrhizal fungi. *Mol. Ecol.* 21, 4160–4170.
- Vellend, M., 2010. Conceptual synthesis in community ecology. *Q. Rev. Biol.* 85, 183–206.
- Wang, J., Shen, J., Wu, Y., Tu, C., Soininen, J., Stegen, J.C., et al., 2013. Phylogenetic beta diversity in bacterial assemblages across ecosystems: deterministic versus stochastic processes. *ISME J.* 7, 1310–1321.
- Wilke, A., Harrison, T., Wilkening, J., et al., 2012. The M5nr: a novel non-redundant database containing protein sequences and annotations from multiple sources and associated tools. *BMC Bioinformatics* 13, 141. <https://doi.org/10.1186/1471-2105-13-141>.
- Yadav, S., Kapley, A., 2019. Exploration of activated sludge resistome using metagenomics. *Sci. Total Environ.* 692, 1155–1164.
- Yang, Y., Woodward, L.A., Li, Q.X., Wang, J., 2014. Concentrations, source and risk assessment of polycyclic aromatic hydrocarbons in soils from midway atoll, north pacific ocean. *PLoS One* 9 (1), e86441. <https://doi.org/10.1371/journal.pone.0086441>.
- Zeng, Q., Liao, C., Terhune, J., et al., 2019. Impacts of florfenicol on the microbiota landscape and resistome as revealed by metagenomic analysis. *Microbiome* 7, 155. <https://doi.org/10.1186/s40168-019-0773-8>.